UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES

THURSDAY, 22 SEPTEMBER 2022 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of FIVE (5) questions .

Answer **ALL** questions in **Section A** and **ONLY ONE(1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

Section A (Answer All Questions)

Q1. (a) A school is planning to organize an outdoor event this Sunday. The weather forecast predicted that there is only 20% chance of bad weather on Sunday. From past records, if the weather is bad, only 30% chance that an event will be successfully organized, but if the weather is not bad, 90% chance that the event will be successful. Let S represents that an event is successful and let B represents that the weather is bad.

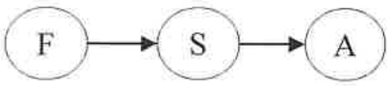
(i) Compute the following probabilities: $P(S|B)$, $P(S|\neg B)$, $P(\neg S|B)$ and $P(\neg S|\neg B)$. (4 marks)

(ii) What is the probability that the weather is bad and the event is successfully organized? (2 marks)

(iii) What is the probability of the event is successful? (3 marks)

(iv) Given that the event is not successful, what is the probability that the weather is bad? (3 marks)

(b) Figure 1.1 shows a Bayesian Network:



F	P(F)
f	0.1
$\neg f$	0.9

F	S	P(S|F)
f	s	0.8
	$\neg s$	0.2
$\neg f$	s	0.05
	$\neg s$	0.95

S	A	P(A|S)
s	a	0.7
	$\neg a$	
$\neg s$	a	
	$\neg a$	0.9

Figure 1.1: Bayesian Network

(i) Write the equation for the joint distribution $P(F, S, A)$. (3 marks)

(ii) Identify the conditional independence relationship in the network. (2 marks)

(iii) Compute the following probabilities: (8 marks)

- $P(f, s, a)$
- $P(s, a)$
- $P(\neg a \mid f, s)$

[Total : 25 marks]

Q2. The following table shows the distance (km) to travel from a particular town to the other. For cases that two towns are not interlinked or there is one-way direction, their distance is marked as "-". For example, the distance from town A to town D is 5km but there is no road going from A to C, E, or F directly (not interlinked), and no road from D back to A (A to D is one-way).

Distance (km)		Town (to)					
		A	B	C	D	E	F
Town (from)	A	0	6	-	5	-	-
	B	6	0	7	-	-	-
	C	-	-	0	-	-	3
	D	-	-	4	0	2	-
	E	-	-	-	-	0	3
	F	-	-	-	-	-	0

(a) If **Breadth-First Search (BFS)** is used to find the path to go from town **A** to **F**,

(i) Draw the search tree generated by the BFS algorithm. If there are several possible nodes to explore, organize and select the node based on *alphabetical* order from left to right. (5 marks)

(ii) What is the path and cost found by BFS to go from A to F? (2 marks)

(iii) Is the path optimal? Justify your answer. (2 marks)

(iv) Given that the branching factor is approximately 2, what is the time complexity to run BFS? (2 marks)

(b) If **Uniform Cost Search (UCS)** is used to find the optimal path to go from town **A** to **F**,

(i) Draw the search tree generated by the UCS algorithm. If there exist nodes with the similar cost, organize and select the node based on *alphabetical* order from left to right. (5 marks)

(ii) What is the path and cost found by UCS to go from A to F? (2 marks)

(iii) Is the path optimal? Justify your answer. (2 marks)

(iv) Given that the branching factor is approximately 2, what is the time complexity to run UCS? (2 marks)

(c) If a **Depth-First Search (DFS)** tree based on alphabetical order is used to find the path to go from town **A** to **F**, is the solution complete? Justify your answer. (3 marks)

[Total : 25 marks]

Q3. (a) Table 3.1 shows the results of a binary classifier.

Table 3.1: Result of Binary Classifier

Actual	**Predicted**
True	True
False	True
True	False
True	True
True	True
True	False
False	False
False	True
True	True
True	False

Determine the following performance measures for the classifier in Table 3.1. Include your calculation in the answer.

(i) Accuracy (2 marks)

(ii) Precision (2 marks)

(iii) Recall (2 marks)

(iv) F_1 score (2 marks)

(b) When designing a classifier for detecting intruders for home security system, between *precision* and *recall*, which performance measure is more critical? Justify your answer. (3 marks)

(c) Figure 3.1 shows the training error and cross-validation error for a certain dataset when varying the model complexity.

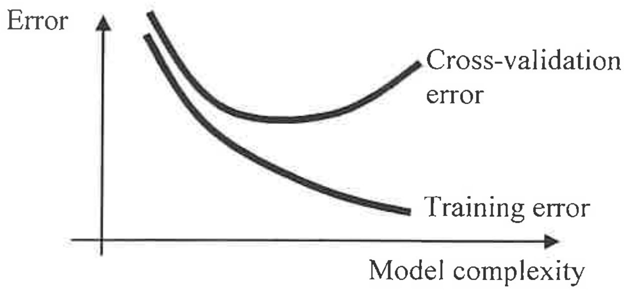


Figure 3.1: Model complexity versus errors.

(i) Copy Figure 3.1 to your answer sheet and identify the region for **high bias** and **high variance**. (2 marks)

Q3. (c) (Continued)

(ii) Explain how are the two phenomena related to model complexity. (4 marks)

(iii) If a model overfitted the data, suggest TWO approaches to resolve the problem. (2 marks)

(d) Assuming binary classifier is used to perform a multiclass classification task with 10 classes by using the following strategies. Compute the total number of binary classifiers needed for each strategy, and identify ONE advantage and ONE disadvantage for each of the strategies.

(i) One-versus-All (OvA) (3 marks)

(ii) One-versus-One (OvO) (3 marks)

[Total : 25 marks]

Section B (Choose Any ONE (1) Question)

Q4. Table 4.1 shows samples extracted from a dataset. The task is to classify y based on x.

Table 4.1

x	y
3	no
5	no
7	yes
9	yes

(a) Suppose **logistic regression** is used to perform the classification.

(i) Given model A ($\theta_0 = -3, \theta_1 = 0.5$) and model B ($\theta_0 = -1, \theta_1 = 0.3$), determine which model is a better fit for the samples. Justify your answer by computing their respective classification accuracies. Show your calculation steps. (9 marks)

(ii) **Gradient descent** is used to find the best parameter to model the samples. Given that the parameter values are initialized to ($\theta_0 = -1, \theta_1 = 0.3$) and learning rate $\eta = 0.01$, compute the updated parameter values after the *first* iteration. Show your calculation steps. (8 marks)

(b) Feature scaling can transform the raw input features into a representation that is more suitable for the Machine Learning algorithm. Perform **Min-max scaling** and **Standardization** separately on x in Table 4.1.

(i) **Min-max scaling**. (3 marks)

(ii) **Standardization**. (5 marks)

[Total : 25 marks]

Q5. Table 5.1 shows samples extracted from a dataset. The task is to predict y values based on x.

Table 5.1

x	y
85	600
80	530
75	300
71	240
69	250

(a) Assume a **linear model** is used to model the samples.

(i) Given two linear models A ($\theta_0 = -1000$, $\theta_1 = 20$) and B ($\theta_0 = -1500$, $\theta_1 = 25$), determine which model is a better fit for the samples based on MSE. Show your calculation steps. (7 marks)

(ii) **Gradient descent** is used to find the best parameter to model the samples. Given that the parameter values are initialized to ($\theta_0 = -1000$, $\theta_1 = 20$) and learning rate $\eta = 0.0001$, compute the updated parameter values after the *first* iteration. Show your calculation steps. (8 marks)

(b) Rather than predicting y values, we want to build a system to classify y into two categories (LOW and HIGH) based on its value.

(i) Convert y in Table 5.1 into two categories, LOW and HIGH, using **equal-distance partitioning**. Show your calculation steps. (4 marks)

(ii) Given a **logistic regression** model with ($\theta_0 = -7.4$, $\theta_1 = 0.1$), compute its classification accuracy on the converted data. (6 marks)

[Total : 25 marks]

Appendix

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP}$$

$$Precision = \frac{TP}{FP + TP}$$

$$F_1 = \frac{TP}{TP + \frac{FN + FP}{2}}$$

$$Recall = \frac{TP}{FN + TP}$$

$$h(\mathbf{x}) = \theta^T \mathbf{x}$$

$$\mathbf{h} = \mathbf{X} \cdot \theta$$

$$MSE(\theta) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$RMSE(\theta) = \sqrt{MSE(\theta)}$$

$$\nabla_\theta MSE(\theta) = \frac{2}{m} \mathbf{X}^T \cdot (\mathbf{X} \cdot \theta - \mathbf{y})$$

$$h_\theta(\boldsymbol{x}) = \sigma(\theta^T \boldsymbol{x}) = \frac{1}{1 + e^{-\theta^T x}}$$

$$\nabla_\theta MSE(\theta) = \frac{1}{m} \mathbf{X}^T \cdot (h_\theta(\boldsymbol{x}) - \mathbf{y})$$

$$P(A, B) = P(A|B)P(B)$$
$$P(B, A) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) + P(\neg A) = 1$$